**25900334** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) Regular EXAMINATION October 2025

## Third SEMESTER

## Discipline Specific Core Course (DSC) - MG3DSCCFS202 - COMPUTER SECURITY

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Explain Clark–Wilson model. [U] / [CO1]
2. Define loss of availability. [K] / [CO1]
3. Define hashing. [K] / [CO1]
4. Name the various filtering techniques used in firewall [K] / [CO1]
5. Name the different types of Intrusion detection system. [U] / [CO1]
6. Explain spyware. [U] / [CO1]
7. Compare the role of symmetric encryption and asymmetric encryption in SSL. [An] / [CO2]
8. Examine how TLS protects against common security threats. [An] / [CO2]
9. Explain one common vulnerability in a web application. [U] / [CO1]
10. Simplify the main difficulty faced in managing keys for database encryption. [An] / [CO2]

11. Summarize the main cause of SQLi attacks. [An] / [CO2]

12. Categorize tools used for static analysis of mobile apps. [An] / [CO2]

**[1x10 = 10]**

## Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain the term access control policy. [U] / [CO1]

14. Distinguish between authentication and authorization. [An] / [CO2]

15. Discuss about the different types of phishing. [U] / [CO1]

16. Write the uses of tunneling protocols in VPN [A] / [CO3]

17. Explain how encryption protects mobile apps. [U] / [CO1]

18. Illustrate the dangers of weak cryptography in web applications. [A] / [CO3]

**[5x4 = 20]**

## Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Discuss the various types of computer security threats. [U] / [CO1]

20. Compare and contrast IPSec Authentication Header (AH) and Encapsulating Security Payload (ESP) in terms of confidentiality, authentication, and integrity [An] / [CO2]

21. Differentiate between cloud network access and traditional IT network access. [An] / [CO2]

**[10x2 = 20]**